



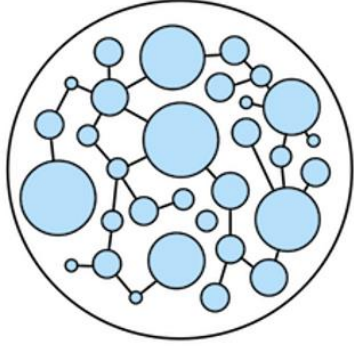
Amazon Neptune **Lecture Slides**

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Good luck on your exam!

What is a Graph database?

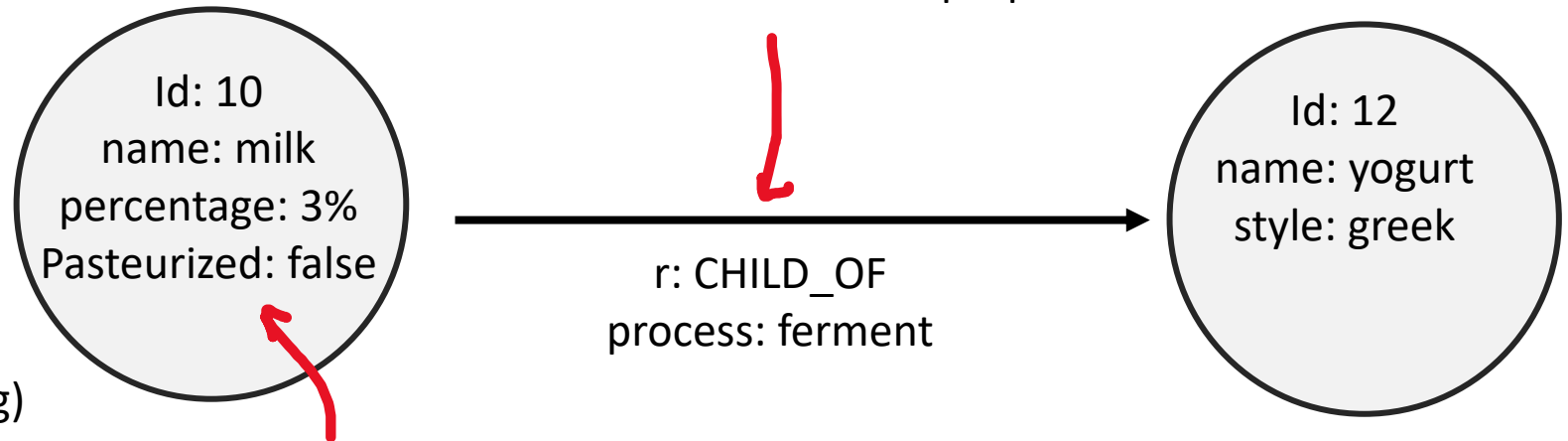


A graph database is a database composed of a data structure that uses vertices (nodes, dots) which form **relationship** to other vertices via edges (arcs, lines)

Use Cases for Graph Database:

- Fraud detection
- Real-time recommendation engines.
- Master data management (MDM)
- Network and IT operations.
- Identity and access management (IAM)
- Traceability in Manufacturing
- Contact Tracing
- Data Lineage for GDPR
- Customer 360-degree analysis (marketing)
- Product recommendations
- Social Media graphing
- Feature Engineering (ML)

The edge can contains **relational data** such as direction and data properties



The node will contain **data properties**

Amazon Neptune



Amazon Neptune is a **graph database** with other offerings around it's graph database.

Amazon Neptune Database

highly available and durable graph database service

- You can use Gremlin, SPARQL or OpenCypher

Amazon Neptune Analytics

Extend Neptune Database by providing capabilities to run large-scale graph analytics algorithms efficiently directly on your operational graph dataset

- load a graph into Neptune Analytics from data in an Amazon S3 bucket or from a Neptune database.
- Integrated with Neptune Workbench
- Includes algorithms like PageRank, shortest path, and community detection directly applicable on the graph data.
- Leverages an Apache Spark cluster to perform complex analytics on graph data.

Amazon Neptune ML

Use graph neural networks (GNNs), a machine learning (ML) technique purpose-built for graphs, to make easy, fast, and more accurate predictions using graph data.

- Powered by the Deep Graph Library (DGL), an open-source library
- Neptune integrates with LangChang

Amazon Neptune Database

Neptune database has two types

- **Neptune Provisioned** — You choose an instance type
- **Neptune Serverless** — A serverless offering. You set a min and max Neptune Capacity Units (NCU)
- Neptune database support Multi-AZ deployment
- Neptune has two storage configuration:
 - Neptune I/O Optimized — increased Input/Output for additional cost
 - Neptune Standard — 25% than I/O Optimized
- You can create a Jupyter Notebook (within Amazon SageMaker Notebook) that includes magic extensions to easily work with you Neptune database.

Neptune Bulk Loader can be used to import large amounts of data

AWS has multiple built-in or third-party options for visualizing your graph database:

- The open-source graph-explorer
- Tom Sawyer Software
- Cambridge Intelligence
- Graphistry
- metaphacts
- G.V()
- Linkurious

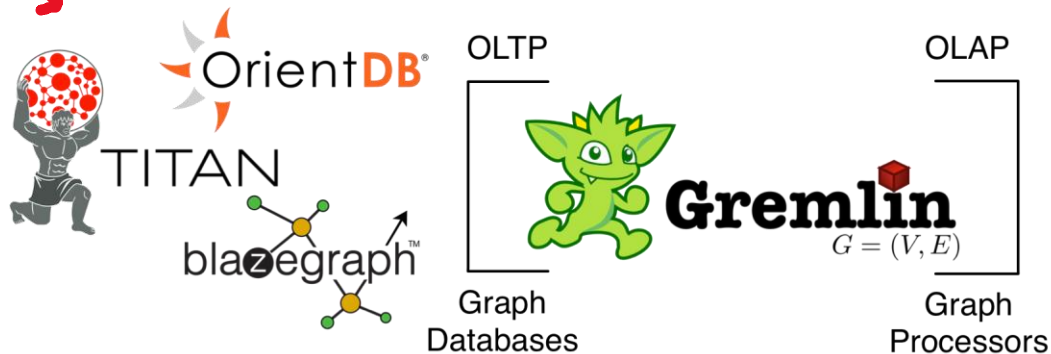
What is Gremlin?

Gremlin is the **graph traversal language** for **Apache TinkerPop**

```
g.V().has("name","gremlin").as("a").  
  out("created").in("created").  
    where(neq("a")).  
  groupCount().by("title")
```

Gremlin is designed to the "Write once, run anywhere" (**WORA**)
Gremlin traversal can be evaluated as

- **real-time database query (OLTP)**
- or as a **batch analytics query. (OLAP)**



Gremlin Host Language Embedding
means You can use your favourite
programming language to write Gremlin

What is OpenCypher?

OpenCypher is an open source implementation of Cypher (the graph query language used by Neo4J)

Cypher is known for being more developer friendly for writing queries than using Gremlin.

Gremlin is generally more proficient at traversal depending on the use-case

```
# Creating
CREATE (a:Person {name: 'Alice', age: 24})
CREATE (b:Person {name: 'Bob', age: 22})
CREATE (a)-[:KNOWS {since: 2021}]->(b)

# Querying
MATCH (p:Person)
RETURN p.name, p.age

MATCH (a:Person {name: 'Alice'})-[:KNOWS]-(b:Person {name: 'Bob'})
RETURN a.name, b.name

# Updating
MATCH (b:Person {name: 'Bob'})
SET b.age = 23
RETURN b.name, b.age

# Advanced Query
MATCH (alice:Person {name: 'Alice'})-[:KNOWS]->(bob:Person)-[:KNOWS]->
(friendsOfBob)
WHERE NOT (alice)-[:KNOWS]->(friendsOfBob)
RETURN friendsOfBob.name AS RecommendedFriend
```

What is SPARQL?

SPARQL (Sparkle) is a Resource Description Framework (RDF) query language. SPARQL allows users to write queries against what can loosely be called "key-value" data or, data that follow the RDF specification of the W3C.

Setting up sample data



```
@prefix : <http://example.org/> .

:Alice a :Person ;
       :name "Alice" ;
       :age 24 ;
       :knows :Bob .

:Bob a :Person ;
     :name "Bob" ;
     :age 22 .
```

Updating data



```
PREFIX : <http://example.org/>
DELETE {
  :Bob :age 22 .
}
INSERT {
  :Bob :age 23 .
}
WHERE {
  :Bob :age 22 .
}
```

Querying data



```
PREFIX : <http://example.org/>
SELECT DISTINCT ?name
WHERE {
  :Alice :knows/:knows ?person .
  ?person :name ?name .
  FILTER NOT EXISTS { :Alice
    :knows ?person }
}
```